CONTRIBUTION STATEMENT

eICU Research Portfolio

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Works covered

1. *Patient Similarity and Model Transportability across eICU hospitals: An outcome-specific analysis of ICU mortality prediction* (research poster).
2. *Patient Similarity and Cross-Hospital Transportability* (software portfolio).

Software implementation and analysis

I was responsible for the project-specific code implementation and integration. My work connected cohort construction, clinical data preparation, and FIDDLE inputs with XGBoost cross-validation and external evaluation. I implemented the project-specific SVD and cosine-similarity analyses, hospital-level summaries, statistical analyses, and figure-generation code. I also carried out the analyses used to develop the poster.

AI coding assistants were used during implementation. I designed the analysis, reviewed the code, and verified the outputs.

Poster writing and presentation

I wrote the poster text and prepared its figures and layout. Keith Feldman, PhD, provided research supervision, discussed the study design and interpretation with me, and suggested revisions to the poster. I was responsible for incorporating the feedback and preparing the revised poster.

Third-party methods and code

The project uses FIDDLE, developed by Tang and colleagues, and adapts the eICU extraction and population workflow released with the FIDDLE experiments. I obtained the label-extraction code and FIDDLE workflow from those public resources. My contribution includes adapting and integrating those components for this project; FIDDLE itself and the upstream code are not my original software contributions.

Relationship between the two works

The poster and software portfolio are complementary artifacts from the same research project. The poster communicates the question, findings, and interpretation; the software portfolio documents implementation and verification. The synthetic demonstration illustrates software behavior and does not constitute an additional clinical study.

Reference: Tang et al. *J Am Med Inform Assoc.* 2020;27(12):1921-1934. doi:10.1093/jamia/ocaa139.
Source repositories: MLD3/FIDDLE and MLD3/FIDDLE-experiments.